# A gamma quotient with strict logarithmic complete monotonicity

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## Task 1. Resolve the gamma-quotient conjecture

Arora and Sahoo, *Interpolation on Gauss hypergeometric functions with an application*, Involve **11**(4) (2018), 625–641, doi:10.2140/involve.2018.11.625, pose Conjecture 3.14 on page 635: for every $0 \le a \le 1$, prove that

$$Q_a(c) = \frac{\Gamma(c)\Gamma(c-4)}{\Gamma(c-a)\Gamma(c+a-1)}$$

decreases on $c > 4$. The answer is affirmative, and strict.

**Theorem 1.** *For every $-3 \le a \le 4$, $c > 4$, and integer $m \ge 1$,*

$$(-1)^m \frac{d^m}{dc^m} \log Q_a(c) > 0.$$

*In particular, $Q_a'(c) < 0$ throughout the conjectured domain, including the parameter endpoints $a = 0, 1$.*

*Proof.* Write $\psi = \Gamma'/\Gamma$. The classical digamma difference formula is

$$\psi(u) - \psi(v) = \int_0^\infty \frac{e^{-vt} - e^{-ut}}{1 - e^{-t}}\, dt, \qquad u, v > 0.$$

For example, it follows by subtraction from DLMF 5.9.16; equivalently, expand $(1 - e^{-t})^{-1}$ and integrate the usual digamma series. All four gamma arguments here are positive. Consequently

$$-\frac{d}{dc} \log Q_a(c) = \int_0^\infty e^{-ct} \frac{1 + e^{4t} - e^{at} - e^{(1-a)t}}{1 - e^{-t}}\, dt. \tag{1}$$

Fix $t > 0$. The function $a \mapsto e^{at} + e^{(1-a)t}$ is convex and takes the same value $e^{-3t} + e^{4t}$ at the endpoints of $[-3, 4]$. It is therefore at most that value throughout the interval. Thus

$$1 + e^{4t} - e^{at} - e^{(1-a)t} \ge 1 - e^{-3t} > 0.$$

At $t = 0$, the numerator in (1) is $3t + O(t^2)$, so the quotient is bounded. At infinity it is $O(e^{4t})$. For $c > 4$, the integral therefore converges absolutely, and it may be differentiated any number of times, locally uniformly in $c$, by exponential domination. For every $m \ge 1$ this gives

$$(-1)^m (\log Q_a)^{(m)}(c) = \int_0^\infty t^{m-1} e^{-ct} \frac{1 + e^{4t} - e^{at} - e^{(1-a)t}}{1 - e^{-t}}\, dt > 0.$$

Finally $Q_a > 0$, so $Q_a' = Q_a (\log Q_a)' < 0$. □

**A short independent proof on the requested range.** The classical series $\psi'(x) = \sum_{k=0}^{\infty}(x+k)^{-2}$ shows that $\psi$ is increasing and strictly concave on $(0,\infty)$ (DLMF 5.15.1). For $0 \le a \le 1$, the concave function $a \mapsto \psi(c-a) + \psi(c+a-1)$ has equal endpoint values. Hence

$$\psi(c-a) + \psi(c+a-1) \ge \psi(c) + \psi(c-1),$$

and therefore

$$(\log Q_a)'(c) \le \psi(c-4) - \psi(c-1) < 0.$$

This also proves the entire original conjecture directly.

**Corollary 2** (Sharp parameter interval)**.** *On the domain where all four gamma arguments are positive,*

$$c > b_a := \max\{4, a, 1-a\},$$

*$Q_a$ is decreasing everywhere if and only if $-3 \le a \le 4$. For precisely these parameters it is strictly logarithmically completely monotone.*

*Proof.* Sufficiency is Theorem 1. If $a > 4$, then $b_a = a$ and $Q_a(c) \to 0$ as $c \downarrow a$, because $\Gamma(c-a) \to \infty$ while the other three factors have finite positive limits. A positive decreasing function cannot have left-endpoint limit zero. If $a < -3$, the same argument at $c = 1-a$ uses $\Gamma(c+a-1) \to \infty$. Thus global decrease fails outside the stated interval, and so does logarithmic complete monotonicity. □

**Status and scope.** The published source retains the conjecture, and the live MathDB record was open with no solution or discussion entries at the dated checks recorded in the accompanying literature audit. No matching full public claim was located in that bounded search. The proof uses classical digamma formulas and elementary convexity; it does not claim those general tools as new.

## Task 2. Find a positive inverse Laplace representation

For each $a \in [-3,4]$, derive an explicit positive measure representing $Q_a(4+x)$ as a Laplace transform. Determine how the measure's density and endpoint behavior depend on $a$.

## Task 3. Determine generalized Stieltjes orders

For each $a \in [-3,4]$, determine the least $\alpha > 0$, if one exists, for which $Q_a(4+x)$ has a representation $\int_0^{\infty}(x+t)^{-\alpha}\,d\mu_a(t)$ with $\mu_a \ge 0$. Include the rational endpoint cases as exact benchmarks.

## Task 4. Sharpen the interpolation error bounds

Use the positive kernel to obtain optimal uniform bounds for $-(\log Q_a)'(c)$ on $0 \le a \le 1$, and propagate these bounds into the quadratic hypergeometric interpolation errors in the source.

## Task 5. Compare decrease with logarithmic complete monotonicity

For quotients of three gamma factors by three gamma factors with fixed total shift difference, find explicit parameter families that are decreasing on their positive-argument domain but fail logarithmic complete monotonicity, or prove their equivalence on a specified family.